INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M3A3 METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M3A3 Cartridge Actuated Thruster Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 3730), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL ORDNANCE STATION (CAGE Codes 19200 and 19203)

10522631	Thruster, Cartridge Actuated, M3A3 Metal Parts Assembly
8796719	Box, Fiberboard (Inner), Packing, Ammunition, For Thruster, Cartridge Actuated, M3A3
8797270	Box, Packing, Ammunition, For Thruster, Cartridge Actuated, M3A3
9217573	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Thruster, Cartridge Actuated, M3A3

U.S. Air Force

TPO 00-953-5567	Thruster, Cartridge Actuated, M3A3

(Application for copies should be addressed to: Commanding Officer, Naval Ordnance Station, Data Control Branch (Code 3720), Indian Head, MD 20640-5035.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to the first article inspection (see 4.3 and 6.3).

3.2 Materials. All materials used in the manufacture of the thrusters shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Raw material. The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.3 Hardness and other physical properties. Hardness and other physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

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3.4 Hydrostatic pressure. The body and end cap shall withstand a hydrostatic pressure of 8,000 pounds per square inch (psi) for 15 seconds, minimum, without leaking, deformation, or mechanical failure (see 6.4.1).

3.5 Structural strength. The assembled thruster shall withstand a static tensile force of 800 pounds, minimum, for 15 seconds, minimum, between mounting points without unlocking and without mechanical failure (see 6.4.1).

3.6 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 41 ± 5 pounds is applied at the steady rate of 0.125 inch per minute.

3.7 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled handtight.

3.8 Protective finish. Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of contour configuration.

3.9 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using procedures in 4.6.6.

3.10 Workmanship. Thrusters shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.10.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.10.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.10.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing

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operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68 to 77•F)
b. Altitude: sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psi

4.4 First article inspection. At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of five metal parts assemblies, nine sets of unassembled parts, one set of unassembled parts without plating or final protective finish, and fifty shear pins from the same lot of shear pins tested by the contractor. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the first article sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new first article samples until an acceptable sample is submitted (see 6.3).

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.5.1.2 Size of lot. The production lot size shall be as specified in the contract.

4.5.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the

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characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.5.2.1 Rod, Piston, Drawing 8594494, covering a detail of Drawing 10522631.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Concentricity of ring groove with large body diameter	Gage
102	Concentricity of body midsection with large body diameter	Gage
103	Concentricity of cavity with pilot diameter	Gage
104	Concentricity of key locking groove with pilot diameter	Gage
105	Concentricity of key escape groove with pilot diameter	Gage
106	Concentricity of pilot with large body diameter	Gage
107	Concentricity of body at rear of ring groove with large body diameter	Gage
108	Diameter of ring groove	Gage
109	Diameter of large body, min	Gage
110	Diameter of body of ring groove	Gage
111	Diameter of body midsection	Gage
112	Diameter of key escape groove, max	Gage
113	Diameter of key locking groove, min	Gage
114	Diameter of pilot, min	Gage
115	Diameter of cavity	Gage
116	Depth of cavity, max	Gage
117	Length from face, pilot end, to end of key escape groove	Gage
118	Length from face, pilot end, to start of key escape groove	Gage
119	Length from face, pilot end, to angle between body and midsection, min	Gage
120	Location of key groove	Gage
121	Profile of key groove	Gage
122	Profile of ring groove	Gage
123	Metal defective	Visual
124	Protective finish inadequate or defective	Visual
125	Surface finish of pilot, key locking groove, and large body improper	Visual
126	Burr or sharp corner	Visual
127	Foreign matter	Visual
Minor		
201	Diameter of large body, max	Gage
202	Diameter of key escape groove, min	Gage
203	Diameter of key locking groove, max	Gage
204	Diameter of pilot, max	Gage
205	Depth of cavity, min	Gage
206	Length of pilot	Gage

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207	Length from face, pilot end, to start of large body diameter, max	Gage
208	Total length	Gage
209	Location of ring groove	Gage
210	Width of ring groove	Gage
211	Marking missing, incorrect, or illegible	Visual
212	Radius, chamfer, or both, missing or incorrect	Visual
213	Surface finish improper, except as otherwise classified	Visual

4.5.2.2 Breech, Assembly, Drawing 8594496, covering a detail of Drawing 10522631.

Critical		
1	Surface finish of cavity improper	Gage
Major		
101	Pitch diameter of large external thread, min	Gage
102	Major diameter of large external thread, min	Gage
103	Pitch diameter of small external thread, min	Gage
104	Major diameter of small external thread, min	Gage
105	Pitch diameter of internal thread, max	Gage
106	Minor diameter of internal thread, max	Gage
107	Pitch diameter of pin hole thread, max	Gage
108	Minor diameter of pin hole thread, max	Gage
109	Concentricity of boss cavity with cavity	Gage
110	Concentricity of striker hole with cavity	Gage
111	Concentricity of pitch diameter of small external thread with cavity Gage	
112	Concentricity of body with pitch diameter of large external thread	Gage
113	Concentricity of pitch diameter of small external thread with pitch diameter of large external thread	Gage
114	Perpendicularity of boss face with pitch diameter of small external thread	Gage
115	Perpendicularity of shoulder with pitch diameter of large external thread	Gage
116	Protrusion of pellets above major diameter of large thread, max Gage	
117	Protrusion of pellets above major diameter of small thread, max Gage	
118	True position of pin hole	Gage
119	Diameter of cavity, max	Gage
120	Diameter of basic location	Gage
121	Diameter of pin hole, max	Gage
122	Depth of counterbore	Gage
123	Depth of thread cavity	Gage
124	Length of effective internal thread, min	Gage
125	Length of internal thread, max	Gage
126	Angle at basic diameter	Gage
127	Height of boss	Gage

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128	Diameter of chamfer, thread	Gage
129	Width of undercut, small external thread, max	Gage
130	Width of undercut, large external thread, max	Gage
131	Angle at mouth of thread cavity	Gage
132	Radius at root of boss	Visual
133	Thread damaged or not full	Visual
134	Metal defective	Visual
135	Protective finish inadequate or defective	Visual
136	Pellet missing, loose, or improperly inserted	Manual/Visual
137	Surface finish of internal thread and mouth chamfer improper	Visual
138	Burr or sharp corner	Visual
139	Foreign matter	Visual
Minor		
201	Pitch diameter large external thread, max	Gage
202	Pitch diameter of small external thread, max	Gage
203	Pitch diameter of internal thread, min	Gage
204	Pitch diameter of pin hole thread, min	Gage
205	Diameter of cavity, min	Gage
206	Diameter of boss cavity	Gage
207	Diameter of pin hole, min	Gage
208	Diameter of body	Gage
209	Diameter of striker hole	Gage
210	Diameter of counterbore	Gage
211	Depth of striker hole	Gage
212	Total length	Gage
213	Length from bottom of cavity to face of boss	Gage
214	Length from large body diameter to shoulder	Gage
215	Location of pin hole	Gage
216	Location of pellets, large thread	Gage
217	Location of pellets, small thread	Gage
218	Diameter of body, large end	Gage
219	Diameter of undercut, large external thread, min	Gage
220	Diameter of undercut, small external thread, min	Gage
221	Diameter of undercut at large body end, min	Gage
222	Diameter of wrench holes	Gage
223	Depth of wrench holes	Gage
224	Length from shoulder to face of boss	Gage
225	Length of undercut and large external thread	Gage
226	Location of wrench holes	Gage
227	Marking missing, incorrect, or illegible	Visual
228	Radius or chamfer incorrect, except as otherwise classified	Gage
229	Surface finish improper, except as otherwise classified	Visual

4.5.2.3 Sleeve, End, Drawing 8594497, covering a detail of Drawing 10522631.

Critical
None defined

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Major		
101	Pitch diameter of thread, min	Gage
102	Major diameter of thread, min	Gage
103	Concentricity of cavity with body	Gage
104	Concentricity of shoulder with body	Gage
105	Symmetry of slot with body	Gage
106	Diameter of body, min	Gage
107	Diameter of cavity, max	Gage
108	Diameter of shoulder, min	Gage
109	Width of slots	Gage
110	Location of slots	Gage
111	Threads damaged or not full	Visual
112	Metal defective	Visual
113	Protective finish inadequate or defective	Visual
114	Surface finish of body improper	Visual
115	Burr or sharp corner	Visual
116	Foreign matter	Visual
Minor		
201	Pitch diameter of thread, max	Gage
202	Diameter of body, max	Gage
203	Diameter of cavity, min	Gage
204	Diameter of shoulder, max	Gage
205	Diameter of thread undercut, min	Gage
206	Depth of cavity	Gage
207	Total length	Gage
208	Length from thread end to flange	Gage
209	Length of slot	Gage
210	Width of shoulder	Gage
211	Location of flats	Gage
212	Location of flange	Gage
213	Width across flats	Gage
214	Width of thread undercut	Gage
215	Angular location of slots	Visual
216	Marking missing, incorrect, or illegible	Visual
217	Radius of chamfer incorrect	Gage
218	Surface finish improper, except as otherwise classified	Visual

4.5.2.4 Key, Drawing 8594498, covering a detail of Drawing 10522631.

Critical
None defined

Major		
101	Radial profile of length	Gage
102	Radial profile of width	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual

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105	Surface finish improper	Visual
106	Burr or sharp corner	Visual
107	Foreign matter	Visual
Minor		
201	Total length	Gage
202	Radius incorrect, except as otherwise classified	Gage

4.5.2.5 Body, Assembly, Drawing 10522633, covering a detail of Drawing 10522631.

Critical
None defined

Major		
101	Pitch diameter of internal thread, max	Gage
102	Minor diameter of internal thread, max	Gage
103	Pitch diameter of external thread, min	Gage
104	Major diameter of external thread, min	Gage
105	Concentricity of small bore with pitch diameter of external thread	Gage
106	Concentricity of pilot diameter with pitch diameter of external thread	Gage
107	Concentricity of large bore with pitch diameter of internal thread	Gage
108	Perpendicularity of face with pitch diameter external thread	Gage
109	Perpendicularity of shoulder with pitch diameter of internal thread	Gage
110	Perpendicularity of small bore with face	Gage
111	Perpendicularity of pilot diameter with face	Gage
112	Diameter of large bore, max	Gage
113	Diameter of small bore, max	Gage
114	Diameter of external thread pilot, min	Gage
115	Depth to bottom of internal thread cavity	Gage
116	Length to shoulder of external thread, min	Gage
117	Length of external thread pilot, min	Gage
118	Diameter of internal thread undercut	Gage
119	Thickness of wall at internal thread undercut, min	Gage
120	Protrusion of pellets, max	Gage
121	Pellets missing, loose, or improperly inserted	Manual/Visual
122	Thread damaged or not full	Visual
123	Angle at bottom of large bore missing or incorrect	Visual
124	Radius or chamfer incorrect	Gage
125	Metal defective	Visual
126	Protective finish inadequate or defective	Visual
127	Surface finish of small bore, pilot, face, and shoulder improper	Visual
128	Burr or sharp corner	Visual
129	Foreign matter	Visual

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Minor

201	Pitch diameter of internal thread, min	Gage
202	Pitch diameter of external thread, max	Gage
203	Diameter of large bore, min	Gage
204	Diameter of small bore, min	Gage
205	Diameter of external thread pilot, max	Gage
206	Diameter of body	Gage
207	Length to shoulder of external thread, max	Gage
208	Length of external thread pilot, max	Gage
209	Total length	Gage
210	Thickness of wall of large bore, min	Gage
211	Diameter of external thread undercut, min	Gage
212	Depth of large bore	Gage
213	Depth of slots	Gage
214	Length of shoulder	Gage
215	Width of slots	Gage
216	Width of internal thread undercut	Gage
217	Width of external thread undercut	Gage
218	Location of marking	Gage
219	Location of pellet	Gage
220	Pellet radius incorrect	Visual
221	Angular location of slots	Visual
222	Marking missing, incorrect, or illegible	Visual
223	Radius or chamfer incorrect, except as otherwise classified	Gage
224	Surface finish improper, except as otherwise classified	Visual

4.5.2.6 Trunnion, Drawing 8594501, covering a detail of Drawing 10522631.

Critical
None defined

Major

101	Concentricity of lugs	Gage
102	True position of lugs with inside diameter	Gage
103	Symmetry of flats at root of lugs with inside diameter	Gage
104	Diameter of lugs, min	Gage
105	Width across flats, min	Gage
106	Thickness of wall, min	Gage
107	Metal defective	Visual
108	Protective finish inadequate or defective	Visual

Minor

201	Diameter of lugs, max	Gage
202	Diameter inside	Gage
203	Total length	Gage
204	Length of lug	Gage
205	Width across flats, max	Gage
206	Thickness of wall, max	Gage
207	Location of lug	Gage

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208	Width of flat Gage	
209	Marking missing, incorrect, or illegible	Visual
210	Radius or chamfer incorrect	Gage
211	Surface finish improper	Visual
212	Burr or sharp corner	Visual
213	Foreign matter	Visual

4.5.2.7 Cap, End, Drawing 10522634, covering a detail of Drawing 10522631.

Critical
None defined

Major

101	Pitch diameter of large thread, max	Gage
102	Minor diameter of large thread, max	Gage
103	Pitch diameter of gas port thread, max	Gage
104	Minor diameter of gas port thread, max	Gage
105	Concentricity of large thread counterbore with pitch diameter of large thread	Gage
106	Concentricity of bore between midsection and key seating cavity with pitch diameter of large thread	Gage
107	Concentricity of key seating cavity with pitch diameter of large thread	Gage
108	Concentricity of sleeve cavity with pitch diameter of large thread	Gage
109	Concentricity of ring groove with sleeve cavity	Gage
110	Concentricity of counterbore, groove end with sleeve cavity	Gage
111	Perpendicularity of large thread counterbore shoulder with pitch diameter of large thread	Gage
112	Perpendicularity of large thread counterbore with counterbore shoulder	Gage
113	Perpendicularity of bore between midsection and key seating cavity with large thread counterbore shoulder	Gage
114	Perpendicularity of key seating cavity with large thread counterbore shoulder	Gage
115	Perpendicularity of sleeve cavity with large thread counterbore shoulder	Gage
116	Diameter of large thread counterbore, max	Gage
117	Diameter of bore between midsection and key seating cavity, min Gage	
118	Diameter of ring groove	Gage
119	Diameter of key seating cavity, max	Gage
120	Diameter of counterbore, groove end	Gage
121	Diameter of sleeve cavity, max	Gage
122	Diameter of thread pilot, gas port	Gage
123	Depth to bottom of large thread counterbore, max	Gage
124	Depth to midsection, cavity	Gage
125	Depth of counterbore, groove end	Gage

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126	Depth of gas port thread pilot	Gage
127	Depth of gas port cavity	Gage
128	Length of effective gas port thread, min	Gage
129	Location of gas port	Gage
130	Diameter of chamfer at mouth of gas port	Gage
131	Diameter of large thread undercut, max	Gage
132	Angle of chamfer at mouth of gas port	Gage
133	Angle of gas port thread chamfer	Gage
134	Width of large thread undercut	Gage
135	Thread damaged or not full	Visual
136	Metal defective	Visual
137	Protective finish inadequate or defective	Visual
138	Surface finish of angles, cavities adjacent to ring groove, sleeve cavity, and face of large thread counterbore improper	Visual
139	Burr or sharp corner	Visual
140	Foreign matter	Visual

Minor

201	Pitch diameter of large thread, min	Gage
202	Pitch diameter of gas port thread, min	Gage
203	Diameter of large thread counterbore, min	Gage
204	Diameter of bore between midsection and key seating cavity, max Gage	
205	Diameter of midsection cavity	Gage
206	Diameter of key seating cavity, min	Gage
207	Diameter of sleeve cavity, min	Gage
208	Diameter of gas port boss	Gage
209	Diameter of body	Gage
210	Depth to bottom of large thread counterbore, min	Gage
211	Depth to bottom of large thread undercut	Gage
212	Depth to bottom of key seating cavity	Gage
213	Depth to key seating cavity	Gage
214	Total length	Gage
215	Location of ring groove	Gage
216	Diameter of gas escape hole	Gage
217	Width of ring groove	Gage
218	Height to top of gas port boss	Gage
219	Profile of ring groove	Visual
220	Marking missing, incorrect, or illegible	Visual
221	Radius or chamfer incorrect, except as otherwise classified	Gage
222	Surface finish improper, except as otherwise classified	Visual

4.5.2.8 Retainer, Cartridge, Drawing 8594503, covering a detail of Drawing 10522631.

Critical
None defined

Major

101	Pitch diameter of thread, max	Gage

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102	Minor diameter of thread, max	Gage
103	Concentricity of body cavity with pitch diameter of thread	Gage
104	Concentricity of head with pitch diameter of thread	Gage
105	Perpendicularity of face of cavity with pitch diameter of thread	Gage
106	Diameter of body cavity, max	Gage
107	Depth of body cavity	Gage
108	Depth of thread cavity	Gage
109	Length of effective thread, min	Gage
110	Internal radius	Gage
111	Number of elongated slots incorrect	Visual
112	Thread damaged or not full	Visual
113	Metal defective	Visual
114	Protective finish inadequate or defective	Visual
115	Surface finish at internal radius, thread cavity, body cavity, and head diameter improper	Visual
116	Burr or sharp corner	Visual
117	Foreign matter	Visual

Minor

201	Pitch diameter of thread, min	Gage
202	Diameter of body	Gage
203	Diameter of body cavity, min	Gage
204	Diameter of head	Gage
205	Drill point in base, max	Gage
206	Total length	Gage
207	Length of head, min	Gage
208	Length to end of external angle, max	Gage
209	Length of elongated slots	Gage
210	Width of elongated slots	Gage
211	Location of elongated slots	Gage
212	Chamfer at threaded end, min	Gage
213	Angular location of slots	Visual
214	Profile of elongated slots	Visual
215	Marking missing, incorrect, or illegible	Visual
216	Radius or chamfer incorrect, except as otherwise classified	Gage
217	Surface finish improper, except as otherwise classified	Visual

4.5.2.9 Spring, Piston, Locking, Drawing 8594504, covering a detail of Drawing 10522631.

Critical
None defined

Major

101	Load at specified heights, incorrect	Gage
102	Solid length, max	Gage
103	Ends correct	Visual
104	Protective finish inadequate or defective	Visual

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105	Burr or sharp edges	Visual
106	Foreign matter	Visual
Minor		
201	Outside diameter	Gage

4.5.2.10 Pin, Firing, Drawing 8594744, covering a detail of Drawing 10522631.

Critical
None defined

Major		
101	Concentricity of groove with body	Gage
102	Concentricity of striker with body	Gage
103	Concentricity of striker shoulder with body	Gage
104	True position of pin hole	Gage
105	Diameter of groove	Gage
106	Diameter of body, min	Gage
107	Diameter of striker, max	Gage
108	Diameter of pin hole, max	Gage
109	Depth of pin hole, max	Gage
110	Profile of striker	Gage
111	Profile of groove	Gage
112	Metal defective	Visual
113	Protective finish inadequate or defective	Visual
114	Surface finish of striker tip and body, improper	Visual
115	Burr or sharp corners	Visual
116	Foreign matter	Visual
Minor		
201	Diameter of body, max	Gage
202	Diameter of striker shoulder	Gage
203	Diameter of striker, min	Gage
204	Length of body to tip of striker	Gage
205	Length of striker	Gage
206	Location of pin hole	Gage
207	Location of groove	Gage
208	Depth of pin hole, min	Gage
209	Depth of slot in base	Gage
210	Total length	Gage
211	Width of groove	Gage
212	Width of slot in base	Gage
213	Location of slot in base	Visual
214	Marking missing, incorrect, or illegible	Visual
215	Radius incorrect	Gage
216	Surface finish improper, except as otherwise classified	Visual

4.5.2.11 Pin, Shear, Drawing 8594506, covering a detail of Drawing 10522631.

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Critical
None defined

Major		
101	Diameter, max	Gage
102	Total length, min	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Surface finish improper	Visual
106	Burr	Visual
107	Foreign matter	Visual
Minor		
201	Total length, max	Gage
202	Profile of end	Gage

4.5.2.12 Assembly (Breech), Drawing 10522631, less all components except Drawings 8594496 and 8594744 and MS28775.

Critical		
1	Firing pin protrusion	Gage
Major		
101	Any part missing	Visual
102	Thread damaged or not full	Visual
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Foreign matter	Visual

Minor
None defined

4.5.2.13 Assembly, Drawing 10522631.

Critical
None defined

Major		
101	Trunnion not free to rotate 360 degrees	Manual
102	End sleeve not free to rotate 360 degrees clockwise	Manual
103	Any part missing, except as otherwise classified	Visual
104	Improper assembly	Visual
105	Thread damaged or not full	Visual
106	Marking missing, incorrect, or illegible	Visual
107	Burr	Visual
108	Foreign matter	Visual

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Minor

201	Protective cap or plug missing	Visual
202	Protective finish inadequate or defective	Visual

4.5.2.14 Box, Fiberboard (Inner), Packing, Ammunition, Drawing 8796719.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual
202	Box damaged	Visual
203	Marking missing, incorrect, or illegible	Visual
204	Cut, damaged, or improperly sealed bag	Visual

4.5.2.15 Box, Packing, Ammunition, Drawing 8797270.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Visual/Manual
202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.5.2.16 Box, Fiberboard (Outer), Alternative, Packing, Ammunition, Drawing 9217573.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual/Visual
202	Carton broken or damaged	Visual
203	Number of units incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

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4.5.3 Inspection testing. The following tests shall be performed in accordance with 4.6.

4.5.3.1 Hardness. Fifty units from each heat treated lot of components shall be tested for compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.5.3.2 Hydrostatic pressure. Each body and end cap shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber subassembly to comply with 3.4 shall be cause for rejection of the subassembly.

4.5.3.3 Structural strength. Each thruster shall be subjected to the structural strength test. Failure of any thruster to comply with the requirements of 3.5 shall be cause for rejection of the thruster.

4.5.3.4 Shear pin strength. Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with 3.6 shall be cause for rejection of the lot.

4.5.3.5 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.5.3.6 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.6.6. Failure of any unit in the inspection sample to comply with 3.9 shall be cause for rejection of the assembly.

4.6 Test methods and procedures.

4.6.1 Hardness. Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the requirements of the applicable drawing.

4.6.2 Hydrostatic pressure. Prior to application of protective finish the body and end cap shall be mounted in the device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.4.

4.6.3 Structural strength. The thruster shall be subjected to the structural strength test in accordance with the requirements of 3.5.

4.6.4 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.6. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.6.5 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.6.6 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and

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moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.7 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 88796719 and 8797270. Level B shall be in accordance with Drawings 88796719 and 9217573.

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-953-5567.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The thruster assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. If first article inspection is required (see 3.1 and 6.3)

d. The responsible inspection activity (see 4.1)

e. Any special instruction requirements (see 4.3)

f. Level of packing and marking (see 5.1)

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors that the items should be a first article sample. The first article should consist of 5 metal parts assemblies, 9 sets of unassembled parts, 1 set of unassembled parts without plating or final protective finish, and 50 shear pins. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4 Definition.

6.4.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.6 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:

Preparing activity
Navy - OS

Navy - OS

Air Force - 99

(Project 1377-0D91)

Review activity:
Air Force - 70

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